

Total No. of Questions : 8]

SEAT No. :

PD4497

[Total No. of Pages : 2

[6403]-305

T.E. (Mechanical/Automobile) (Honors)
E-VEHICLE TECHNOLOGY
(2019 Pattern) (Semester - V) (302031MJ)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw the neat sketch wherever necessary.

- Q1)** a) Write a short note on different components of li-ion battery with neat sketch? **[4]**
- b) What are the different chemistries of li-ion batteries? Mention in short significance of each of them for electric vehicle. **[8]**
- c) Explain the advantages and disadvantages and applications of li-ion batteries. **[8]**

OR

- Q2)** a) Explain the following parameters of li-ion batteries in short (any two), **[4]**
- i) Cell voltage
 - ii) Energy density
 - iii) Power density
- b) What is the type of charging li-ion battery? Explain the precautions needs to be taken while charging. **[8]**
- c) Write a short note on Availability of lithium ion batteries and government policies to fulfill the demands of lithium batteries for Indian e-vehicles. **[8]**

- Q3)** a) Explain battery management system with working mechanism and neat sketch? **[8]**
- b) Explain lead acid batteries with advantages, disadvantages and applications. **[8]**

OR

P.T.O.

Q4) a) Write a short note on need of thermal management system in e-vehicles? Explain with mechanism. **[8]**

b) Explain Li-ion supercapaciter with advantages, disadvantages and applications. **[8]**

Q5) a) Write a short note on Mechanical and electrical connections of motors. **[8]**

b) List out the Output characteristics of electric motor drives for EVs? **[8]**

OR

Q6) a) Write a short note on Output Characteristics of Motor Drives in EVs? **[8]**

b) Explain different types of drives used in electric vehicle with neat sketches. **[8]**

Q7) a) Explain the significance of implementation of IOT in electric vehicle on basis of wireless sensor network with neat sketch wherever required. **[10]**

b) What is battery swapping? Explain with Advantages And Challenges of Battery Swapping. **[8]**

OR

Q8) a) Explain the Significance of IOT for modes of fast and efficient charging in electric vehicles. **[8]**

b) Write a short note on Battery Standards in electric vehicles with any two types? **[10]**

x

x

x